

CNN Architectures for Image Classification

MNIST, CIFAR-10 & a channel-attention residual variant

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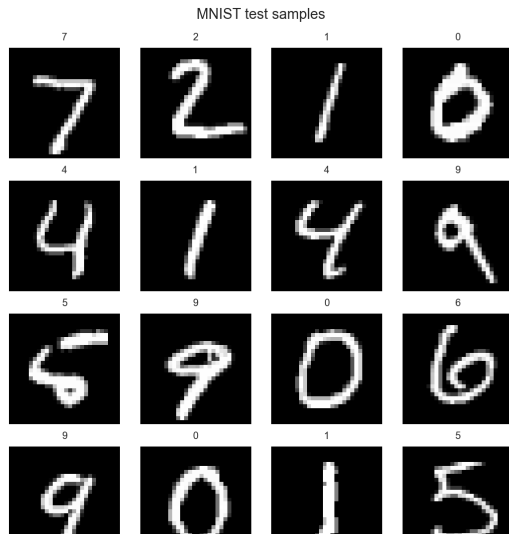
Problem, datasets, and goal

Task. 10-class image classification.

Datasets.

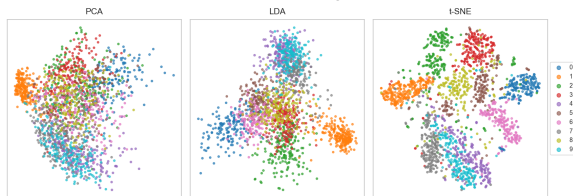
- **MNIST** – handwritten digits, 28×28 grayscale, 60k train / 10k test, 10 classes (0–9).
- **CIFAR-10** – natural objects, 32×32 RGB, 50k train / 10k test, 10 classes (airplane, car, bird, cat, deer, dog, frog, horse, ship, truck).

Goal. Compare seven CNN architectures (one is our own modification of ResNet-18) under a single training pipeline using accuracy, macro precision/recall/F1, and AUROC.

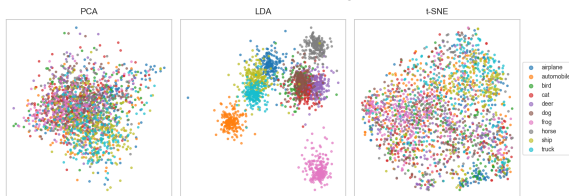


Class structure: PCA $\rightarrow$ LDA $\rightarrow$ t-SNE

MNIST raw pixels



CIFAR-10 raw pixels



Takeaway. MNIST is already well-separable in low-dim projections of the raw pixels (LDA in particular). CIFAR-10 classes overlap heavily in pixel space — this is why a flat MLP plateaus near 50%, while a CNN can recover separable feature representations.

Models compared, and our modification

Seven architectures, identical training recipe.

Model	Family
MLP	baseline, no spatial bias
SimpleCNN	2 conv + 2 FC
LeNet-5	classic 1998 CNN
AlexNet*	5 conv + 3 FC
VGG-11*	deep 3×3 + BatchNorm
ResNet-18*	residual, 4 stages
SE-ResNet-Lite	ours

* Adapted for 32×32 inputs.

Our novel architecture — **SE-ResNet-Lite**.

- Take ResNet-18 as the backbone.
- Insert a **Squeeze-and-Excitation block** inside every residual block. SE = global-pool $\rightarrow$ 2-layer MLP $\rightarrow$ sigmoid $\rightarrow$ per-channel gating.
- Replace the stem with a **depthwise-separable** 3×3 DW $\rightarrow$ 1×1 PW.
- Net parameter overhead vs ResNet-18: $< 1\%$.

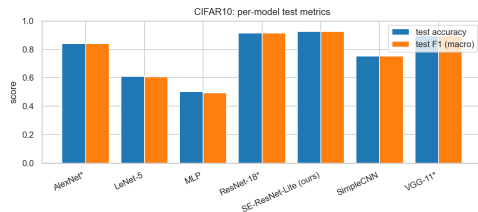
Hypothesis. Channel attention boosts accuracy; the lighter stem keeps the parameter budget essentially fixed.

Results and conclusions

Model	MNIST acc	CIFAR-10 acc	F1
MLP	99.03%	50.18%	0.493
SimpleCNN	99.46%	75.34%	0.752
LeNet-5	99.31%	61.02%	0.606
AlexNet	99.57%	84.03%	0.840
VGG-11	99.62%	89.48%	0.895
ResNet-18	99.65%	91.38%	0.914
SE-ResNet-Lite	99.70%	92.67%	0.927

Findings.

- CNNs dominate MLP on CIFAR-10 by > 30 pp; on MNIST, all models are within 1 pp.
- Residual connections beat VGG-11 at a smaller parameter budget.
- SE-ResNet-Lite is the best model on *both* datasets, at $< 1\%$ extra parameters vs ResNet-18.



One limitation. Single seed, no hyperparameter sweep.

One future improvement. Train ablations of SE-only and DW-stem-only, evaluate on Tiny ImageNet, sweep the SE reduction ratio.